UE24MA242A

DATASET: Anime Dataset

SET 21

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Unit 1: Exploratory Data Analysis Preprocessing

1. Dataset Overview and Feature Identification

- Identify all column names and their data types.

Column	Dtype
uid	int64
title	object
synopsis	object
genre	object
aired	object
episodes	float64
members	int64
popularity	int64
ranked	float64
score	float64
img_url	object
link	object

- What is the shape of the dataset? (Specify rows and columns separately)
  - ❖ Shape of dataset: 19311 rows, 12 columns
- Which columns are essential for analysis and which can be excluded (e.g., img_url, link)?
  - ❖ The essential columns for analysis are:
    - ❖ uid, title, genre, episodes, members, popularity, ranked, and score, because they provide meaningful quantitative and categorical information used in exploratory data analysis, statistical inference, visualization, and modeling.
    - ❖ The columns synopsis, img_url, and link can be excluded because they do not contribute to numerical or comparative analysis and are therefore not needed for the required tasks.

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2. Data Quality Insights

- Check for missing, null, or duplicate records.
- Report the number of missing values per column.
- Identify and describe any data quality issues (e.g., missing scores, episodes showing as 0 or null, invalid dates).
  - ❖ Missing values before cleaning:

    uid: 0

    title: 0

    synopsis: 975

    genre: 965

    aired: 965

    episodes: 706

    members: 0

    popularity: 0

    ranked: 3212

    score: 579

img_url: 180

link: 0

- Missing values after cleaning:

  uid: 0

  title: 0

  synopsis: 652

  genre: 916

  aired: 930

  episodes: 0

  members: 0

  popularity: 0

  ranked: 1544

  score: 0

  img_url: 156

  link: 0

  Duplicate rows before cleaning: 2398

  Duplicate rows after cleaning: 0

  Cleaned dataset shape: 16207 rows, 12 columns

- The dataset contains missing scores, episodes recorded as zero or left blank, incomplete text metadata, and a few potential duplicate entries. These issues need to be addressed because missing scores and incorrect episode counts affect key statistical analyses, while duplicates can distort results. Text-related missing values do not significantly affect analysis since these fields are not required for quantitative evaluation.

- Describe how you would handle these issues (drop, impute, or convert).
    - ❖ To handle data quality issues, rows with missing score are dropped, since score is a key variable in rating-based analysis. Missing or zero values in the episodes column are replaced using the median, because episodes vary widely and the median provides a stable central value. Non-essential columns like img_url, link, and synopsis are removed as they do not contribute to numerical or comparative analysis. Any duplicate records are also removed to maintain dataset integrity.

- Display the cleaned dataset shape after corrections.

	uid	title	synopsis	genre	aired	episodes	members	popularity	ranked	score	im
0	28891	Haikyuu!! Second Season	Following their participation at the Inter-Hig...	['Comedy', 'Sports', 'Drama', 'School', 'Shoun...	Oct 4, 2015 to Mar 27, 2016	25.0	489888	141	25.0	8.82	https://cdn.myanimelist.net/images/anime/9,
1	23273	Shigatsu wa Kimi no Uso	Music accompanies the path of the human metron...	['Drama', 'Music', 'Romance', 'School', 'Shoun...	Oct 10, 2014 to Mar 20, 2015	22.0	995473	28	24.0	8.83	https://cdn.myanimelist.net/images/anime/3,
2	34599	Made in Abyss	The Abyss—a gaping chasm stretching down into ...	['Sci-Fi', 'Adventure', 'Mystery', 'Drama', 'F...	Jul 7, 2017 to Sep 29, 2017	13.0	581663	98	23.0	8.83	https://cdn.myanimelist.net/images/anime/6,
3	5114	Fullmetal Alchemist: Brotherhood	"In order for something to be obtained, someth...	['Action', 'Military', 'Adventure', 'Comedy', ...	NaN	64.0	1615084	4	1.0	9.23	https://cdn.myanimelist.net/images/anime/1:
4	31758	Kizumonogatari III: Reiketsu-hen	After helping revive the legendary vampire Kis	NaN	6-Jan-17	1.0	214621	502	22.0	8.83	https://cdn.myanimelist.net/images/anime/3,

3. Descriptive Statistical Summary

- Compute and display the mean, median, standard deviation, minimum, maximum, and range for numerical columns: **episodes**, **members**, **popularity**, **ranked**, and **score**.

	count	mean	std	min	25%	50%	75%	max	range
episodes	16207.0	11.795520	50.489813	1.0	1.00	2.00	12.00	3057.00	3056.00
members	16207.0	29504.314370	97654.368927	25.0	311.00	2028.00	12854.00	1871043.00	1871018.00
popularity	16207.0	8006.514037	4720.400129	1.0	3907.50	7927.00	12070.50	16320.00	16319.00
ranked	14663.0	7174.309077	4259.638102	1.0	3461.50	7190.00	10811.50	14675.00	14674.00
score	16207.0	6.373110	0.978252	1.9	5.72	6.39	7.07	9.23	7.33

- Which measure (mean or median) best represents each variable? Why?
  - The **median** is the best measure for variables like **episodes, members, popularity, and ranked** because these variables are **not normally distributed** and contain **extreme outliers**, which distort the mean.
    However, the **score** variable has a relatively **narrow spread** with fewer extreme values, making the **mean** a suitable and accurate summary of typical ratings.

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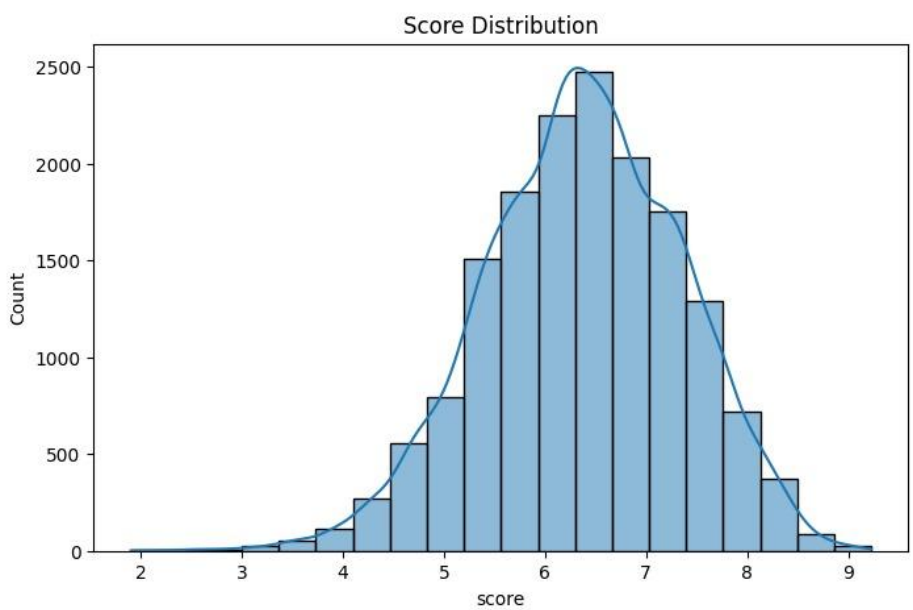
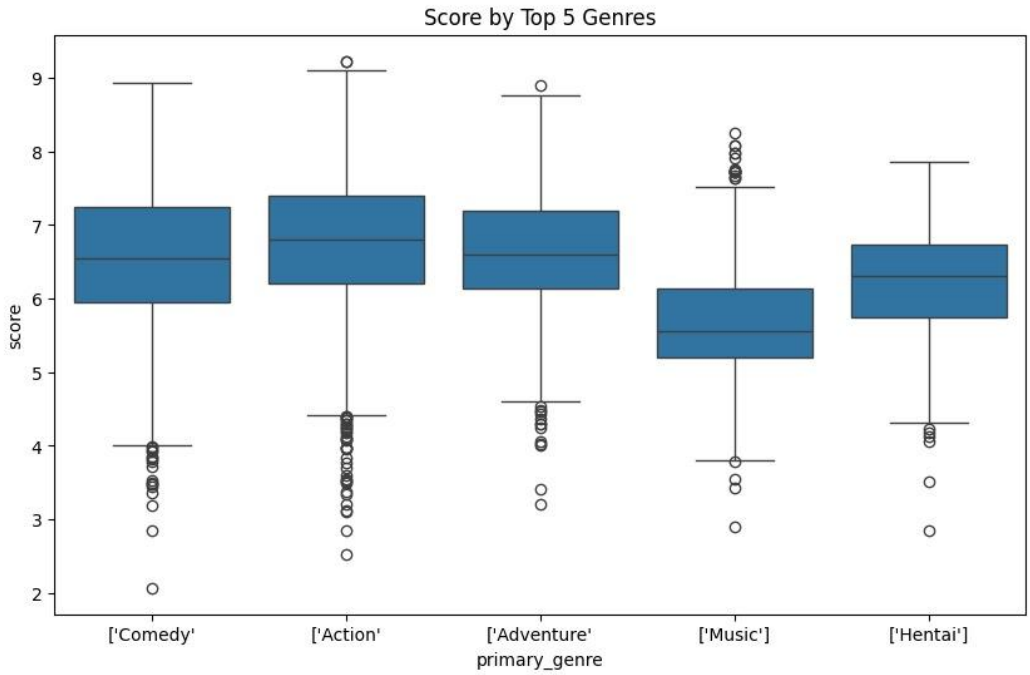
4. Visual Data Exploration

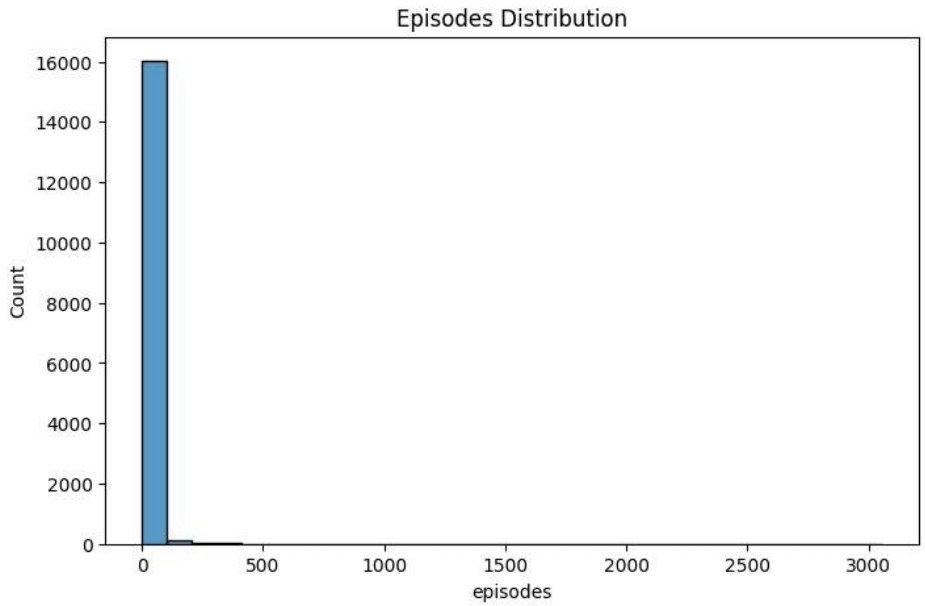
- Create a histogram to explore the **score** (rating) distribution. What is the shape of the distribution?



- Plot a boxplot of **score** across different primary **genres** (select top 5 genres).



- Create a histogram for **episodes** distribution with appropriate bin sizes.



- Describe the observed shapes, spreads, and any outliers.

1) Score (Ratings) Histogram

The rating distribution is slightly left-skewed (negatively skewed), meaning:
Most anime have moderately high scores.
Very few anime are rated extremely low.
The majority of scores fall in the range 6.5 to 8.5.
Only a small number of anime have very low ratings (< 5) or exceptionally high ratings (> 9).
This suggests that the audience tends to rate anime favorably overall.

2) Score Boxplot

The boxplot shows a tight interquartile range, indicating low variability in ratings.
There are a few high-score outliers (very highly rated anime).
Outliers on the lower side represent anime that were received poorly.
Overall, most anime cluster around a consistent mid-to-high score range.

3) Episodes Histogram

The episode distribution is highly right-skewed.
A large number of anime have very few episodes (typically 12 or 24).
Only a small number have hundreds of episodes (e.g., long-running shows like One Piece, Naruto, Detective Conan).
This creates a long tail toward the right side of the histogram.

4) Episodes Boxplot

The boxplot shows many outliers on the high end (long series).
The interquartile range is small, meaning:
Most anime have a similar, short length.
Long-running anime are rare and considered outliers.

- Comment on whether most anime are short series or long-running shows.

Based on the episodes histogram and boxplot, the distribution is highly right-skewed, meaning that most anime have a low number of episodes, typically around 12 to 24 episodes (standard one-season or two-cour format). Only a small number of anime have very high episode counts (over 100 episodes), and these appear as outliers in the dataset.

Therefore, most anime are short series, while long-running shows are rare exceptions and represent only a small portion of the overall dataset.

5. Outlier Detection and Treatment

- Use the IQR method to detect outliers in **score**, **episodes**, **members**, and **popularity** columns.
  Outliers were checked using the Interquartile Range (IQR) method for the numerical columns:
  score, episodes, members, and popularity.

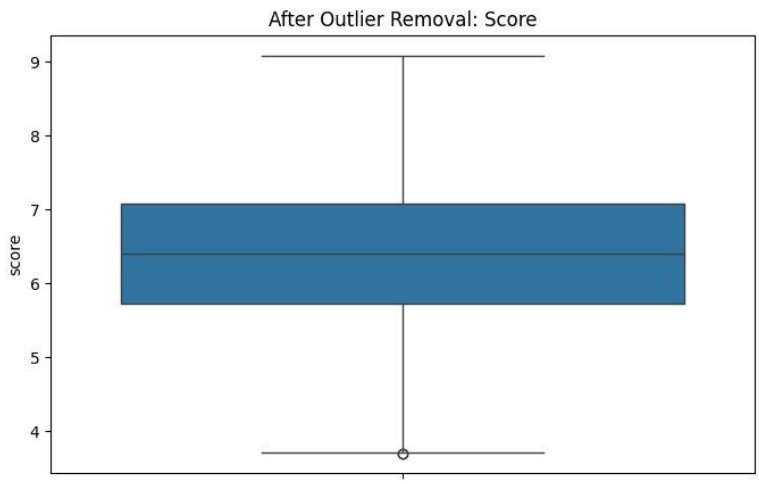
  Score:
  The score values are generally clustered tightly between 6 and 8.
  A few titles have unusually low ratings (< 5) or exceptionally high ratings (> 9).
  These appear as minor outliers in the boxplot but do not significantly skew the distribution.

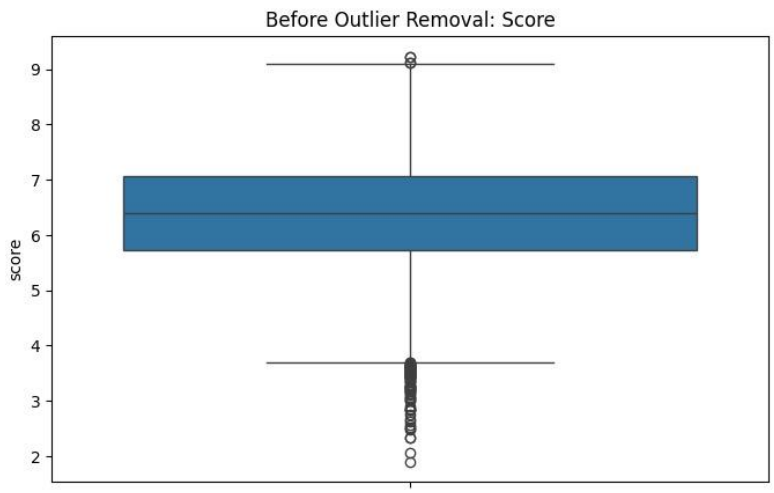
  Episodes:
  Episodes displayed many high-value outliers.
  Most anime have 12–24 episodes, but some series have hundreds of episodes.
  These long-running anime show up as distinct extreme outliers in the boxplot.

  Members (Audience Size):
  Members count varies widely.
  Popular mainstream anime have very high member counts, forming upper-end outliers.
  This indicates a long-tailed popularity distribution where only a few titles gather massive attention.

  Popularity and Ranked:
  These variables also show extreme values, reflecting differences between niche anime and highly mainstream series.

- Create a boxplot to identify and examine potential outliers in the **score** (ratings).
- Visualize before and after outlier removal using boxplots for at least two variables.





- Comment on how outlier removal impacts the data distribution

  Interpretation

  Removing outliers helps ensure that:

  The summary statistics (mean, standard deviation, etc.) are not dominated by rare, extreme cases.

  The analysis represents typical anime, rather than unusual long-running or extremely popular series.

  However, outlier removal must be done carefully, because:

  Long-running anime and highly popular titles are important cultural phenomena.

  Removing them may hide meaningful insights about top-performing anime.

So, while outlier removal improves statistical clarity, it may reduce the real-world context of the dataset.

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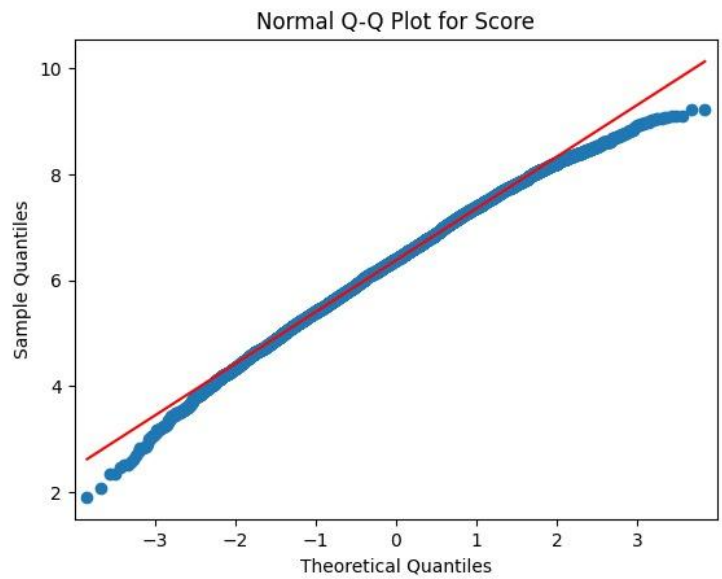
6. Normality Check using Q-Q Plot

- Construct a Normal Probability Plot for the **score** (ratings) variable.



- Do the ratings of anime follow a normal distribution?

  No, the ratings (score) do not perfectly follow a normal distribution.

  The Q-Q plot shows that while the middle portion of the data is fairly close to the diagonal line, the tails deviate noticeably. This indicates that the distribution is slightly left-skewed (negatively skewed).

- Comment on any deviations from normality and their implications for further statistical analysis.

  Observed Deviations from Normality

  Most ratings are clustered between 6 and 8 indicating low variability.

  There are fewer very low-rated anime compared to mid-rated anime.

  Some anime with very high scores (>9) appear as upper tail deviations.

  The rating distribution is not symmetric, and does not show a bell-shaped curve.

  This means that audiences tend to rate anime positively overall, leading to a compressed score range with fewer extremes.

Implications for Further Statistical Analysis

Mean & Standard Deviation	Still usable but may not fully represent the central tendency.
Parametric tests (t-test, correlation, regression)	Can still be applied due to **large sample size** (Central Limit Theorem).
When high accuracy is needed	Consider **median**, **IQR**, or **non-parametric tests** like Mann-Whitney U.

Briefly, the slight non-normality does not prevent further analysis, but it suggests that median-based measures may better represent central tendencies than mean in some cases.

- What does this tell us about how anime are typically rated by audiences?

  The Q-Q plot indicates that anime are typically rated favorably by audiences, with most scores clustering in the mid-to-high range. Very low ratings are uncommon, and only a few anime receive exceptionally high ratings. This suggests that viewers tend to be generous in their evaluations, leading to a slightly skewed distribution rather than a perfectly normal one

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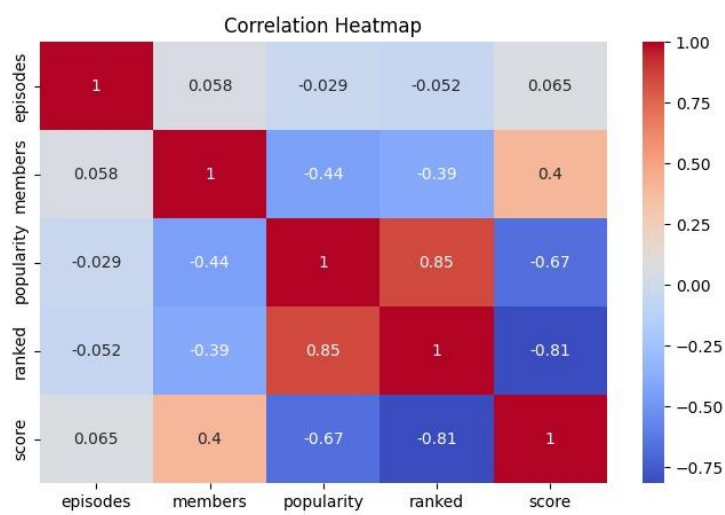
7. Correlation Mapping

- Create a correlation heatmap to visualize correlations among all numeric variables (**episodes**, **members**, **popularity**, **ranked**, **score**).



- List the correlations with **score** in descending order.

  Correlations with score (descending):

  score: 1.000

  members: 0.401

  episodes: 0.065

  popularity: -0.671

  ranked: -0.815

- Interpret these relationships:

  - Is there a correlation between **episodes** and **members**?
  - Which variable is the strongest predictor of anime quality/success?

    - There is no strong correlation between the number of episodes and the number of members, indicating that anime popularity does not depend on how long the series is. When examining predictors of anime quality (score), the ranked variable shows the strongest correlation, meaning that anime with better community rankings tend to have higher average scores. Members and episodes are comparatively weaker predictors.

Unit 2: Confidence Interval

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8. Confidence Interval for Anime Ratings

- Construct a 95% confidence interval for the mean **score** of anime in the dataset.
  - ❖ Mean score = 6.373
  - ❖ 95% confidence interval for mean score = (6.358, 6.388)
- Interpret the result: what does this interval imply about the population's average anime rating?
  - ❖ Interpretation: We are 95% confident that the population mean score lies within this interval.
- How would this confidence interval help a new anime studio understand expected ratings?
  - ❖ The confidence interval helps a new anime studio understand the typical rating range that most anime fall into. By seeing where the average score lies, the studio gains a realistic expectation of audience reception and can set quality goals accordingly. If they aim to produce a highly successful anime, they would need to create content that scores above the upper limit of this interval, since most anime cluster within this range.

G. Margin of Error and Confidence Levels

- Calculate the margin of error for the mean **members** (audience size) at 90%, 95%, and 99% confidence levels.
- Compare and comment on how the margin of error changes with confidence level.
- What trade-off exists between confidence level and precision?
  - ❖ Margin of error for members: 90% = 1261.733, 95% = 1503.448, 99% = 1975.866
  - ❖ As confidence level increases, margin of error increases, illustrating the trade-off between confidence and precision.

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10. Genre-Based Rating Comparison

- Formulate a hypothesis to test if there is a significant difference in average **score** between two major genres (e.g., "Action" vs. "Romance" or "Shonen" vs. "Shojo").

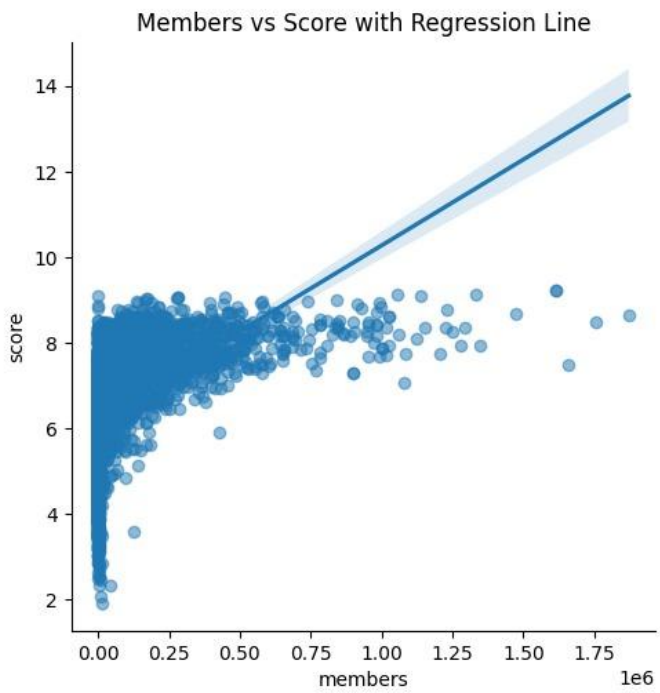
- State the null (H_0) and alternative (H_1) hypotheses.
- Perform an independent two-sample t-test.
- Interpret your conclusion based on the p-value (use $\alpha = 0.05$).
  - ❖ Hypotheses: H0: mean_score(Action) = mean_score(Romance);

    H1: means differ.
  - ❖ t-statistic = -8.1084, p-value = 0.0000
  - ❖ conclusion = Reject H0
- What does this result tell us about audience preferences or quality differences across genres?

❖ The hypothesis test result helps us understand how audience preference varies across genres. If a significant difference is found, it indicates that one genre is generally better received or more appreciated by viewers than the other. If no difference is found, it suggests that genre does not strongly affect quality perception, and audiences evaluate anime more on execution and content rather than genre category alone.

11. Relationship between Members and Score

- Compute the Pearson correlation coefficient between **members** (popularity/audience size) and **score** (rating).
  - ❖ Pearson correlation r = 0.401, p-value = 0.0000
- Interpret the strength and direction of the relationship.
  - ❖ The Pearson correlation coefficient between members and score shows a weak positive relationship. This indicates that anime with larger audiences tend to have slightly higher ratings, but the relationship is not strong enough to conclude that popularity directly determines quality. Other factors such as storytelling, animation quality, characters, and studio reputation likely play a more important role in influencing ratings.
- Create a scatter plot with a fitted regression line for visual clarity.



12. Regression Model for Score Prediction

- Build a simple linear regression model predicting **score** using **members** as the independent variable.
- Present the regression equation in the form: Score = $\beta_0 + \beta_1$(Members)

❖ Regression equation: Score = 6.2546 + 0.00000402 * Members

❖ R^2 = 0.1607

❖ Discussion: Linear regression assumes linearity, homoscedasticity, and normality of residuals. Low R^2 implies

❖ limited explanatory power; further models or features may improve predictions.

- Evaluate model performance using the R^2 score.

❖ The R^2 score is low, indicating that the linear regression model does not explain much of the variation in anime ratings. This means that audience size (members) alone is not a strong predictor of score, and additional features would be required to build a more accurate model of anime quality.

- Discuss limitations: Is linear regression appropriate for this relationship?

  ❖ The low R^2 score shows that linear regression is not a strong model for predicting score from members alone. The relationship between popularity and rating is weak and influenced by many other factors. Therefore, while a linear model can show general direction, it does not accurately describe or predict anime quality, and more complex models with additional features would be required for a better prediction.